ECSE 331
Laboratory No. 4
Design of a BJT Amplifier in PSPICE

April 9, 2021

Seyed Yazdan Zinati
260928077
seyed.zinati@mail.mcgill.ca

James Wu
260929944
james.wu4@mail.mcgill.ca

***Abstract*—**

I. Introduction

II. Analysis

A. Theoretical Design

We need to find resistor values for R_1, R_2, R_{sig}, R_E, and R_C in Figure 1.

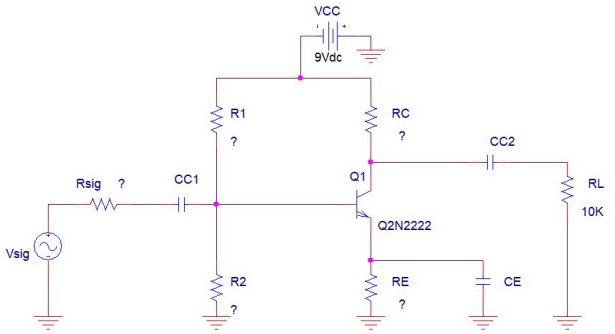


Fig. 1. Design template circuit schematic

1) Small Signal Model (AC Analysis):

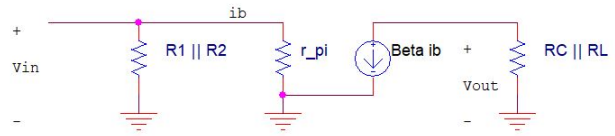


Fig. 2. Small signal model (AC Analysis) circuit

From the small signal model we can deduce:

$$Rin = R_1||R_2||r_\pi \tag{1}$$

$r_\pi = {}^{V_T}/_{I_B}$ and $V_T = 25 \times 10^{-3}V$ therefore:

$$I_B = \frac{V_T}{r_\pi} = \frac{25 \times 10^{-3}}{r_\pi} \tag{2}$$

Write V_{out} as a function of i_b and i_b as a function of V_{in}:

$$V_{out} = -\beta i_b (R_C||R_L) \tag{3}$$

$$i_b = {}^{V_{in}}/_{r_\pi} \tag{4}$$

From (3) and (4) we can write V_{out} as follows:

$$V_{out} = -\beta \frac{V_{in}}{r_\pi}(R_C||R_L) \tag{5}$$

Rewriting Equation (5) we get the following:

$$\frac{V_{out}}{V_{in}} = \left|\frac{-\beta}{r_\pi}(R_C||R_L)\right| \Rightarrow \frac{V_{out}}{V_{in}} = \frac{\beta}{r_\pi}(R_C||R_L) \tag{6}$$

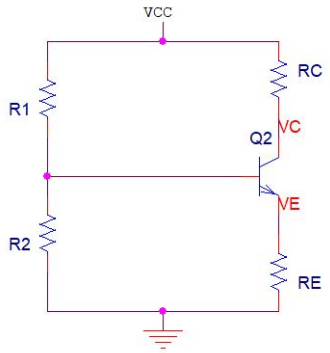


Fig. 3. DC Analysis circuit

2) (Small Signal Model) DC Analysis:

$$V_B - V_E = 0.7V \tag{7}$$

$$\frac{V_{CC} - V_B}{R_1} = I_B + \frac{V_B}{R_2} \Rightarrow V_B = \frac{-I_B R_1 R_2 - V_{CC} R_2}{R_1 + R_2} \tag{8}$$

Choosing the following parameters: $\begin{cases} R_1 = R_2 = 10K\Omega \\ R_{sig} = 1K\Omega \\ \beta = 100 \end{cases}$

and From Equation (1) we find r_π:

$$R_{in} = R_1||R_2||r_\pi = 1K\Omega \Rightarrow r_\pi = 1250\Omega$$

From Equation (2) we find I_B:

$$I_B = \frac{25 \times 10^{-3}}{1250} = 2 \times 10^{-5}A$$

Finding R_C using (6) and knowing that $R_L = 10K\Omega$:

$$\frac{V_{out}}{V_{in}} = \frac{100(R_C||R_L)}{r_\pi} = 50 \Rightarrow \boxed{R_C = 666.67\Omega}$$

Using Equation (8) with $V_{CC} = 9V$, we find $V_B = 4.4V$ so from (7), we get $V_E = 3.7V$.

$$I_E = (\beta + 1)I_B \Rightarrow I_E = 2.02 \times 10^{-3}A$$

$$R_E = \frac{V_E}{I_E} \Rightarrow \boxed{R_E = 1831.68\Omega}$$

Resistor values to be used in our design:

$$\begin{cases} R_1 = 10K\Omega \\ R_2 = 10K\Omega \\ R_{sig} = 1K\Omega \\ RC = 666.67\Omega \\ R_E = 1831.68\Omega \end{cases}$$

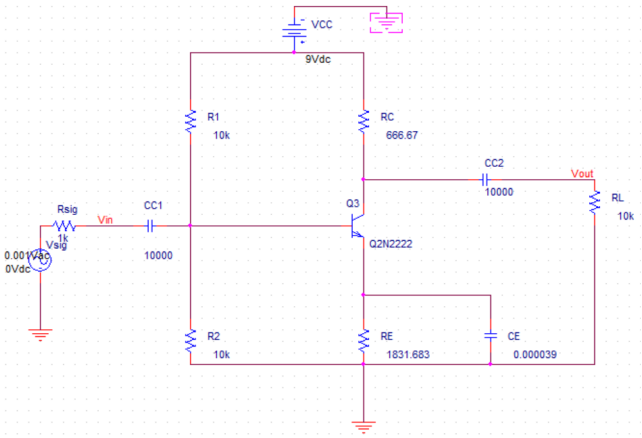


Fig. 4. BJT Amplifier design circuit schematic

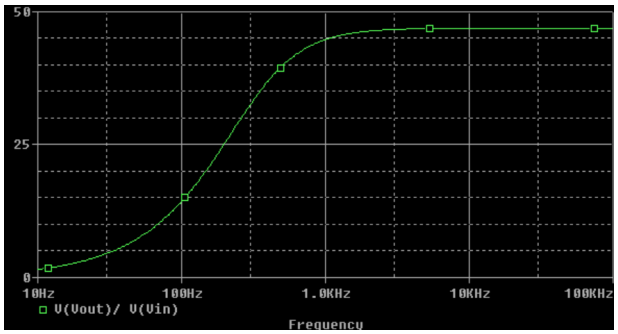


Fig. 5. BJT Amplifier V_{out}/V_{in} as a function of frequency

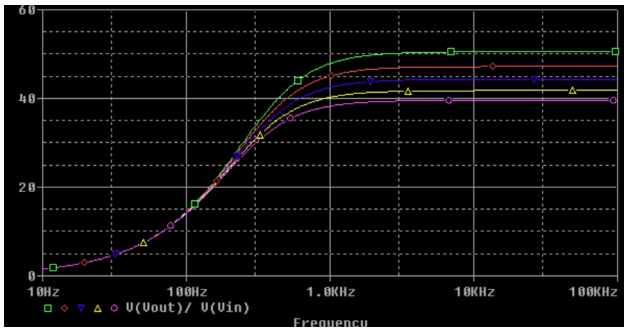


Fig. 6. BJT Amplifier V_{out}/V_{in} as a function of frequency at 0°C, 25°C, 50°C, 75°C, and 100°C

B. Implementation of the design in PSPICE

III. Conclusion

IV. Appendices